

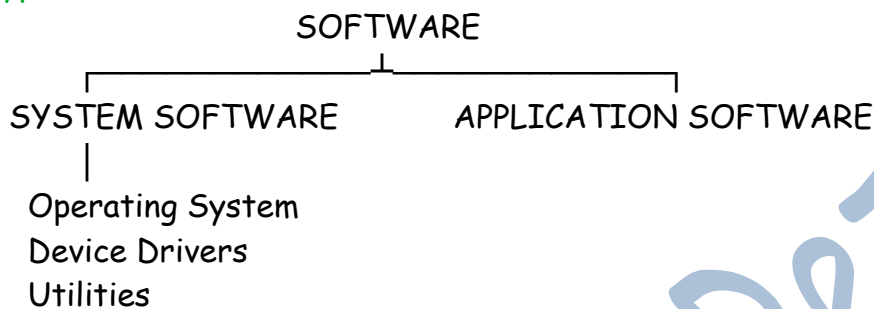
UNIT : II

Software & its types :-

What is Software?

Software is a set of programs, instructions and related data that tells a computer how to perform specific tasks. It controls and manages the operations of computer hardware and enables users to interact with and use the computer system effectively.

Types Of Software



FIRMWARE → separate category/type

1. System Software:

System software is a set of programs that controls and manages the computer hardware and provides a platform for running application software. It acts as an interface between the user, application software, and computer hardware.

Functions of System Software

1. **Manages hardware** - Controls and coordinates hardware devices such as CPU, memory, keyboard, printer, etc.
2. **Manages system resources** - Allocates CPU, memory and other resources to different programs.
3. **Provides a platform** - Provides an environment in which application software can run.
4. **Controls input and output** - Manages communication between the computer and input/output devices.
5. **Maintains system operation** - Helps the computer operate efficiently and reliably.

Types of System Software

- **Operating Systems:** It manages computer hardware, system resources, files, and provides an environment for running application software. Windows, Linux, macOS
- **Device Drivers:** They allow the operating system to communicate with and control hardware devices such as printers, keyboards, and scanners.
- **Utility Programs:** It performs maintenance tasks such as disk management, file management, backup, security, and system optimization.
- **Communication Software:** It enables computers and devices to exchange data and communicate through networks and communication systems.
- **Programming Language Translators:** They convert programs written in high-level or assembly languages into machine language that the computer can understand and execute.

2. Application Software:

Application software is a set of programs designed to perform specific tasks according to the needs and requirements of the user. It helps users perform various activities such as writing, calculations, communication, designing, and entertainment.

Functions Of Application Software

- **Document Creation** - Helps users create, edit, format, and print documents.
- **Data Management** - Helps users store, organize, and manage data efficiently.
- **Calculations** - Performs mathematical calculations and analyzes numerical data.
- **Graphics and Designing** - Helps users create, edit, and modify images and designs.
- **Communication** - Helps users communicate and exchange information through networks and the Internet.
- **Education and Learning** - Provides lessons, tutorials, exercises, and other learning materials.

Types of Application Software

1. Word Processing Software:

Word processing software is used to create, edit, format, save, and print text documents.

Examples: MS Word, WordPad, Google Docs.

2. Spreadsheet Software

Spreadsheet software is used to organize data in rows and columns, perform calculations, and prepare charts.

Examples: MS Excel, Google Sheets.

3. Database Software

Database software is used to store, organize, manage, and retrieve large amounts of data efficiently.

Examples: MS Access, MySQL, Oracle.

4. Graphics Software

Graphics software is used to create, edit, and manipulate pictures, drawings, designs, and other graphical objects.

Examples: CorelDRAW, Adobe Photoshop, Paint.

5. Personal Assistant Software

Personal assistant software helps users perform everyday tasks such as setting reminders, searching information, and managing schedules.

Examples: Siri, Google Assistant, Microsoft Copilot.

6. Educational Software

Educational software is designed to support teaching and learning by providing lessons, tutorials, exercises, tests, and interactive learning materials.

Examples: Educational apps, learning platforms, digital tutorials.

7. Entertainment Software

Entertainment software is used for enjoyment and leisure activities such as watching videos, playing games, and listening to music.

Examples: Media players, video games, streaming applications.

Firmware Software

Firmware is a special type of software stored permanently in the memory of a hardware device. It provides instructions to control and manage the basic operations and functions of that device.

Functions of Firmware

1. **Controls Hardware** - Controls the basic operations and functions of a hardware device.
2. **Starts the Device** - Helps hardware perform its initial operations when the device is switched on.
3. **Controls Device Functions** - Provides instructions that tell the device how to perform its basic tasks.
4. **Provides Hardware Communication** - Helps hardware components communicate and work correctly with other system components.

Examples

- **Printer:** Controls printing mechanisms and paper movement.
- **Router:** Controls network and data communication functions.
- **Keyboard:** Controls how key presses are detected and sent to the computer.

Operating System & its Functions

Operating System (OS)

An Operating System (OS) is a system software that acts as an interface between the user and computer hardware. It manages hardware resources, controls system operations, and provides an environment for running application software.

Examples: Windows, Linux, macOS, Android, iOS.

Functions OF Operating System:

1. Process Management

The OS manages running programs and processes. It allocates CPU time to different processes and controls their execution.

2. Memory Management

It manages the computer's main memory (RAM). It allocates memory to programs when required and releases it after use.

3. File Management

The OS manages files and folders. It allows users to create, delete, copy, move, rename, and organize files.

4. Device Management

It controls input and output devices such as keyboards, printers, mice, and scanners through device drivers.

5. User Interface

The OS provides an interface through which users interact with the computer. It may be a **Graphical User Interface (GUI)** or **Command-Line Interface (CLI)**.

6. Security and Protection

It protects system resources and user data by providing features such as passwords, user accounts, and access permissions.

7. Input and Output Management

It manages the communication between the computer and its input/output devices and controls the flow of data.

8. Resource Management

It manages important computer resources such as the **CPU, memory, storage, and peripheral devices** and allocates them efficiently.

9. Error Detection and Handling

The OS detects system errors and takes appropriate actions to prevent problems and maintain smooth system operation.

Types of Operating System

1. Single-User Operating System

A Single-User Operating System is designed to allow **only one user to use the computer system at a time**. It is mainly used on personal computers where one person works on the system.

Example: MS-DOS, Windows for personal use.

2. Multi-User Operating System

A Multi-User Operating System allows **multiple users to access the same computer system or its resources at the same time**. It manages the resources so that different users can work independently.

Examples: UNIX, Linux.

3. Multitasking Operating System

A Multitasking Operating System allows a user to **run and manage more than one task or application at the same time**. The operating system switches the CPU between tasks to provide smooth operation.

Example: Windows, Linux, macOS.

4. Batch Operating System

A Batch Operating System collects **similar jobs and processes them together in groups or batches**. The jobs are processed automatically without requiring continuous interaction from the user.

Example: Payroll processing and electricity bill processing.

5. Real-Time Operating System (RTOS)

A Real-Time Operating System processes data and gives a **response within a specific time limit**. It is used where quick and timely responses are very important.

Examples: Air traffic control systems, industrial control systems, and medical equipment.

Computer Languages

A **computer language** is a language used to communicate instructions to a computer. It helps programmers write programs that tell the computer **what tasks to perform and how to perform them**.

Computer languages are mainly classified into the following three types:

1. Machine Language

Machine language is the **lowest-level computer language** and consists of instructions written in the form of **binary digits (0 and 1)**. It is directly understood and executed by the computer without any translator. However, it is difficult for humans to write and understand.

Example: 10110000 01100001

Merits

- It is **directly understood by the computer**.
- It does not require a translator for execution.
- It provides **fast execution** because instructions are in machine code.
- It can make efficient use of computer resources.

Demerits

- It is **very difficult for humans to understand and write**.

- It is difficult to find and correct errors.
- Programs are **machine-dependent**.
- Writing large programs in machine language is very time-consuming.

2. Assembly Language

Assembly language is a **low-level language** that uses symbolic codes or **mnemonics** instead of long binary instructions. It is easier to understand and write than machine language. An **assembler** is used to convert assembly language into machine language.

Examples: ADD, SUB, MOV

Merits

- It is **easier to understand** than machine language.
- It uses mnemonic codes such as **ADD, SUB, and MOV**.
- It provides better control over computer hardware.
- Programs can be executed efficiently.

Demerits

- It is more difficult than high-level languages.
- It is **machine-dependent**.
- An **assembler** is required to convert it into machine language.
- Developing large programs takes more time.

3. High-Level Language

High-level languages are designed to be **easy for humans to read, write, and understand**. They use English-like words and mathematical symbols. A compiler or interpreter is required to translate them into machine language.

Examples: C, C++, Java, Python.

Merits

- It is **easy to learn, write, and understand**.
- Programs are easier to **debug and maintain**.
- It reduces the time required for program development.
- Many high-level languages are **portable** and can work on different computer systems with suitable translators.

Demerits

- It requires a **compiler or interpreter** for translation.
- It generally provides less direct control over hardware.
- Execution can be slower than machine language in some cases.
- Programs may require more memory and system resources.

Translator

A **translator** is a system software that converts a program written in a programming language into machine language. It makes the program understandable and executable by the computer. **Compiler, Interpreter and Assembler** are types of translators.

1. Compiler

A **compiler** is a translator that converts the **entire high-level language program** into machine or object code at once. It reports errors after analyzing the program. The compiled program can then be executed.

2. Interpreter

An **interpreter** is a translator that translates and executes a high-level language program **one statement at a time**. It stops when it finds an error. It generally does not produce a separate object code.

3. Linker

A **linker** is a system program that combines different **object files and library files** to form a single executable program. It also connects the required functions and references of different program modules.

4. Loader

A **loader** is a system program that loads an **executable program from secondary storage into main memory (RAM)**. It allocates the required memory and prepares the program for execution.

Computer Virus

A **computer virus** is a malicious program that can copy itself and spread from one computer or file to another. It can damage files, corrupt data, slow down the computer, and affect the normal working of the system.

Main Characteristics / Effects of Computer Virus

1. **Self-Replication:** A virus can make copies of itself and spread to other files or computers.
2. **File Damage:** It may corrupt, modify, or delete important files.
3. **Slow Performance:** It can make the computer slow by using system resources.
4. **Data Loss:** It may cause loss or damage to important data.
5. **Spreading:** It can spread through infected files, USB drives, email attachments, downloads, etc.

Prevention

- Do not open unknown or suspicious files.
- Scan USB drives and downloaded files.
- Keep the operating system and security software updated.
- Use reliable antivirus software.

Computer Antivirus

Antivirus software is a program designed to detect, prevent, and remove computer viruses and other harmful software from a computer. It protects the computer and its data from virus attacks.

Main Characteristics / Functions

1. **Virus Detection:** It identifies viruses and other harmful programs present in the computer.
2. **Virus Removal:** It removes or quarantines infected files from the system.
3. **Real-Time Protection:** It continuously monitors the computer and detects suspicious activities.
4. **File Scanning:** It scans files, programs and storage devices for viruses.
5. **Protection:** It helps protect the computer and important data from virus infections.

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